Problem Sheet: Week 9

1. In order to test $H_0 : \mu = -1$ v. $H_1 : \mu = +1$, a single x is taken from N$(\mu, 1)$.

   a) Show that the Bayesian decision, regardless of the losses and the priors, is "H_0 if $x < k$, H_1 if $x > k$, either if $x = k$" for some value k.

   b) Obtain expressions for the probabilities of the type I and type II errors in terms of k, using the N$(0, 1)$ c.d.f. $\Phi(z)$.

   c) With losses $L_{01} = 2$, $L_{10} = 1$ and priors $p_0 = \frac{1}{3}, p_1 = \frac{2}{3}$, determine the Bayesian decision and compare it with the Neyman-Pearson decision with level 0.05.

2. Let X be a single observation from Binomial$(4, p)$ and let the prior distribution be $p \sim$Beta$(3, 1)$. For the test $H_0 : p \leq \frac{1}{2}$ v. $H_1 : p > \frac{1}{2}$

   a) find the prior probability that H_0 is true

   b) find the posterior probability that H_0 is true given $x = 1$

   c) with linear losses: $L_1(p) = 2(p - \frac{1}{2})$ for $p > \frac{1}{2}$ and $L_2(p) = (\frac{1}{2} - p)$ for $p \leq \frac{1}{2}$, find the Bayesian decision given $x = 1$ (i.e., that which minimises the expected loss).